SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Somaiya TRUST

Module 5
Knowledge Representation and Reasoning

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The world is everything that is the case.

Wittgenstein, Tractatus

Knowledge Based Agents

- Human intelligence is not only reflex-based; it uses **reasoning over internal knowledge**.
- In AI, this approach to intelligence is embodied in **knowledge-based agents**.
- A **Knowledge-Based Agent** is an intelligent agent that:
  - Stores knowledge about the world
  - Uses reasoning to derive new knowledge
  - Takes decisions based on inferred information

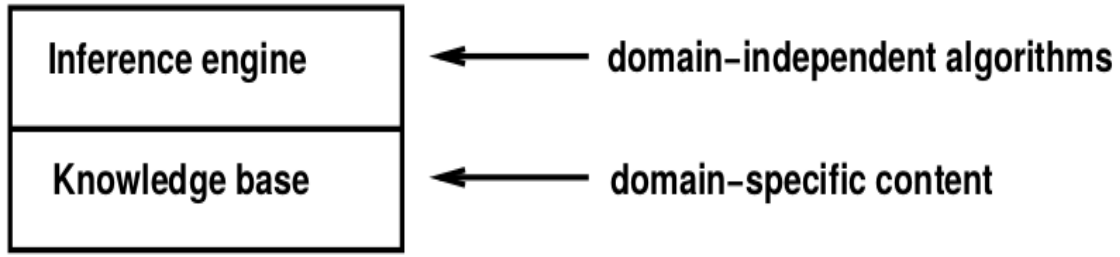
Knowledge Base Agents

- A **knowledge base (KB)** is ***a set of sentences in a formal language***.
- **Inference**: Deriving new sentences from old sentences in KB.

- Each time ***a knowledge-based agent*** does three things.
  - **TELL**
    - Adds new information (percepts) to KB
  - **ASK**
    - Queries KB to decide actions
  - **Inference**
    - Derives new sentences from existing knowledge

Structure of Knowledge Base Agent

- The agent must be able to
  - Represent states and actions
  - Incorporate new precepts
  - Update internal representations of the world
  - Deduce hidden properties of world
  - Deduce appropriate action



A Simple Knowledge-Based Agent

```
function KB-AGENT(percept) returns an action
  static: KB, a knowledge base
          t, a counter, initially 0, indicating time

  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))
  action ← ASK(KB, MAKE-ACTION-QUERY(t))
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))
  t ← t + 1
  return action
```

- MAKE-PERCEPT-SENTENCE constructs a sentence asserting that the agent perceived the given percept at the given time.
- MAKE-ACTION-QUERY constructs a sentence that asks what action should be done at the current time.
- MAKE-ACTION-SENTENCE constructs a sentence asserting that the chosen action was executed.
- The details of the inference mechanisms are hidden inside TELL and ASK.

Logic in General

- Knowledge base consist of sentences.
- Sentences are expressed according to the **syntax of logic**.
- E.g.
- “ X + Y = 4”
  - a well-formed sentence,
- “x4y+=
  - is not.
- The **semantics** defines the **truth** of each sentence with respect to each **possible world (model).**

Logic in General (Contd.)

- For the sentence “X + Y = 4” where 0 < X,Y <= 3
- The possible models are
  - All possible assignments of numbers to the variables X and Y.

- (1,1), (1,2), (1,3), (2,1),(2,2),(2,3),(3,1), (3,2) and (3,3)
- The sentence is true in a world where
  - X is 1 and Y is 3,
  - X is 2 and Y is 2 and
  - X is 3 and Y is 1
  - But false in all other cases

Logic in General(Contd.)

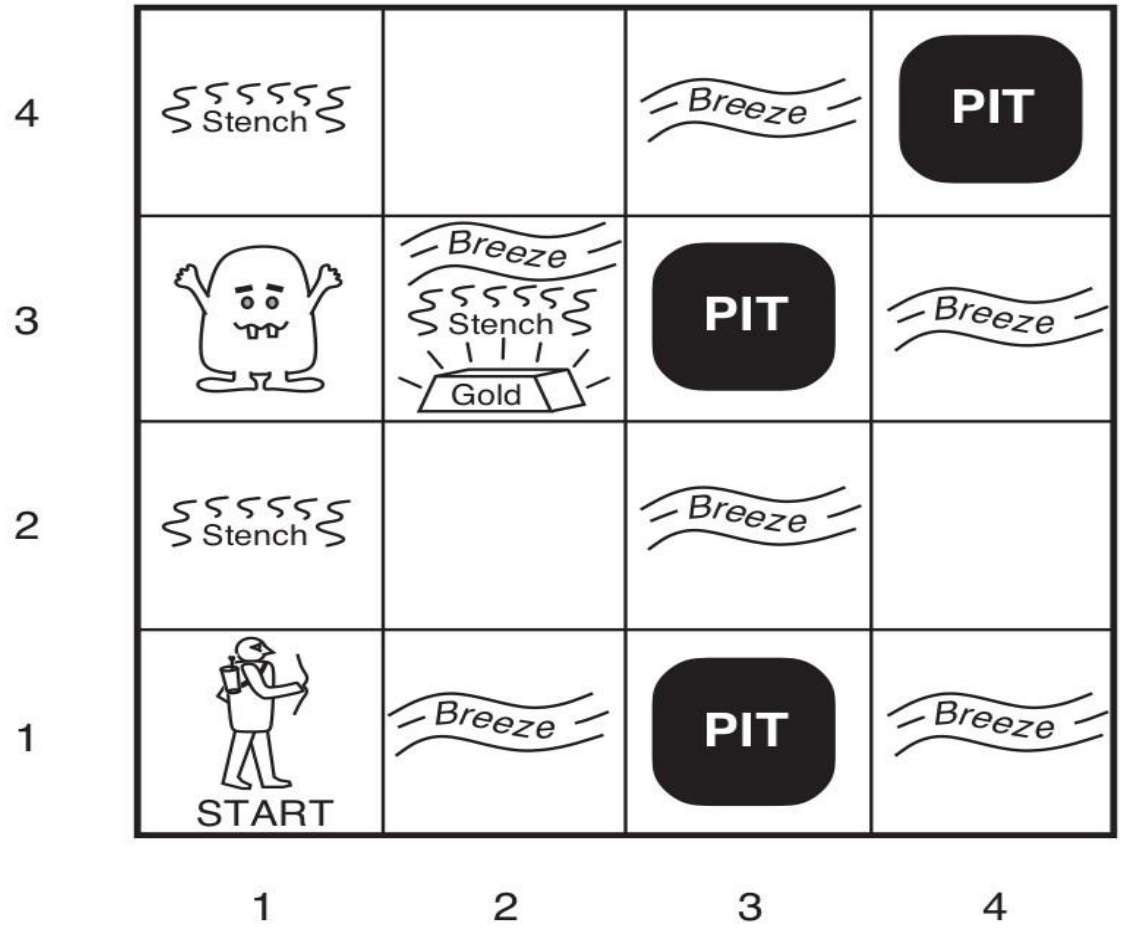
- Logical entailment between sentence is that a sentence follows logically from another sentence.
- Mathematically $\alpha \vDash \beta$
- If and only if, in every model in which α is true, β is also true

$$\alpha \vDash \beta \text{ if and only if } M(\alpha) \subseteq M(\beta)$$

- E.g.
  - α : x + y = 4
  - β : x * y >= 2
  - For all x,y <=3

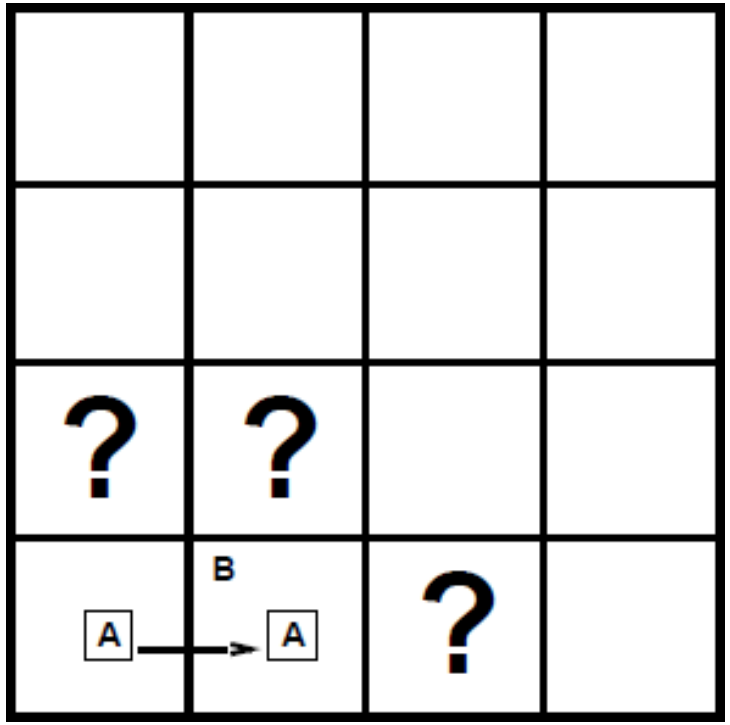
Entailment in Wumpus World

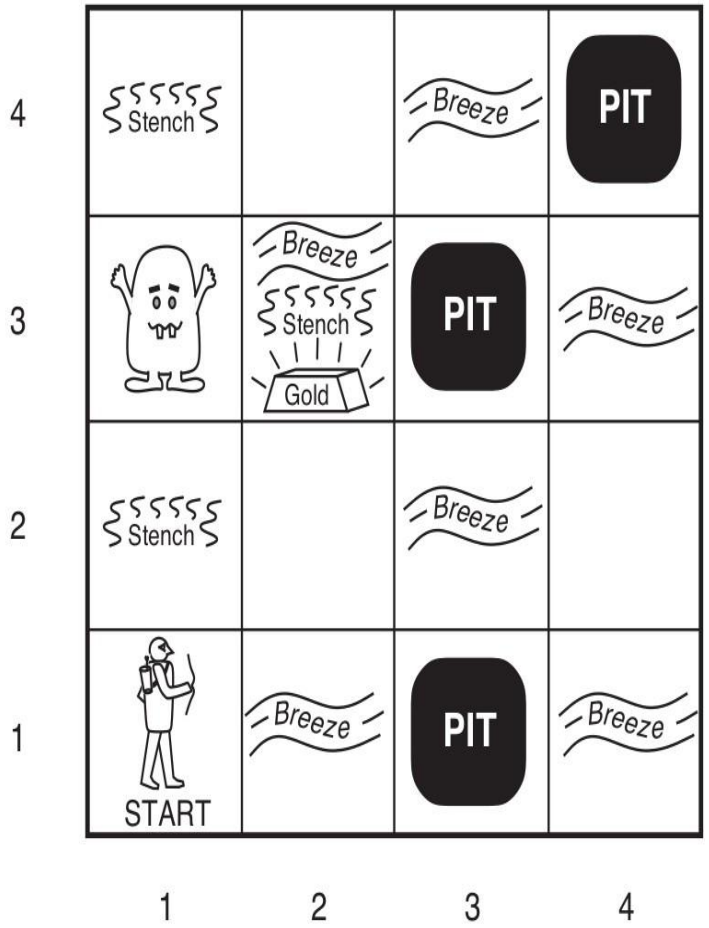
- Consider the scenario shown in the figure the agent



Entailment in Wumpus World (Contd.)

- Situation after detecting nothing in [1,1], moving right, breeze in [2,1]
- The agent is interested in whether the adjacent squares [1,2],[2,1] and [3,1] contain pits.



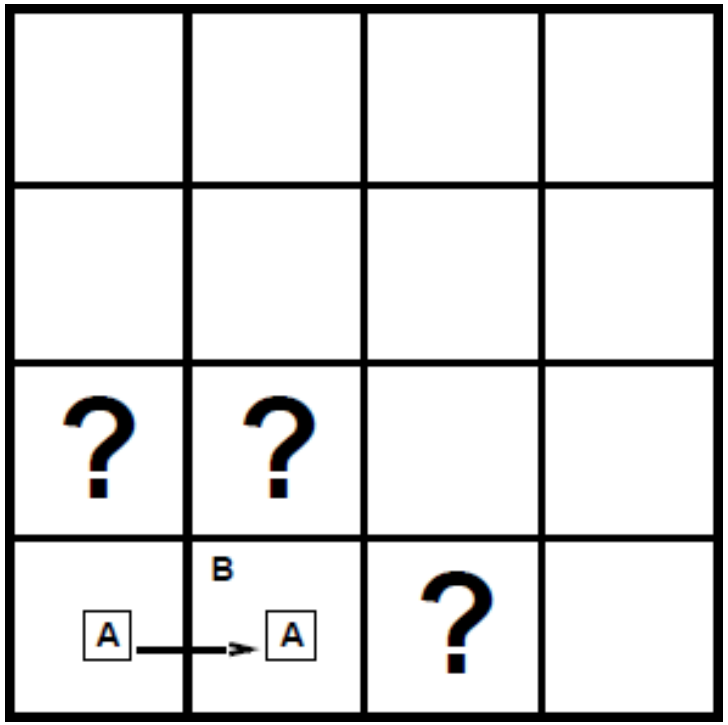


Entailment in Wumpus World

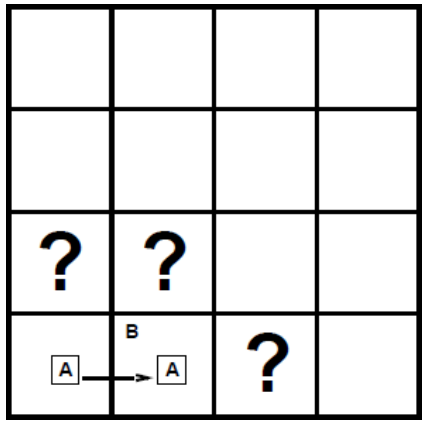
Each of three squares might or might not contain a pit.

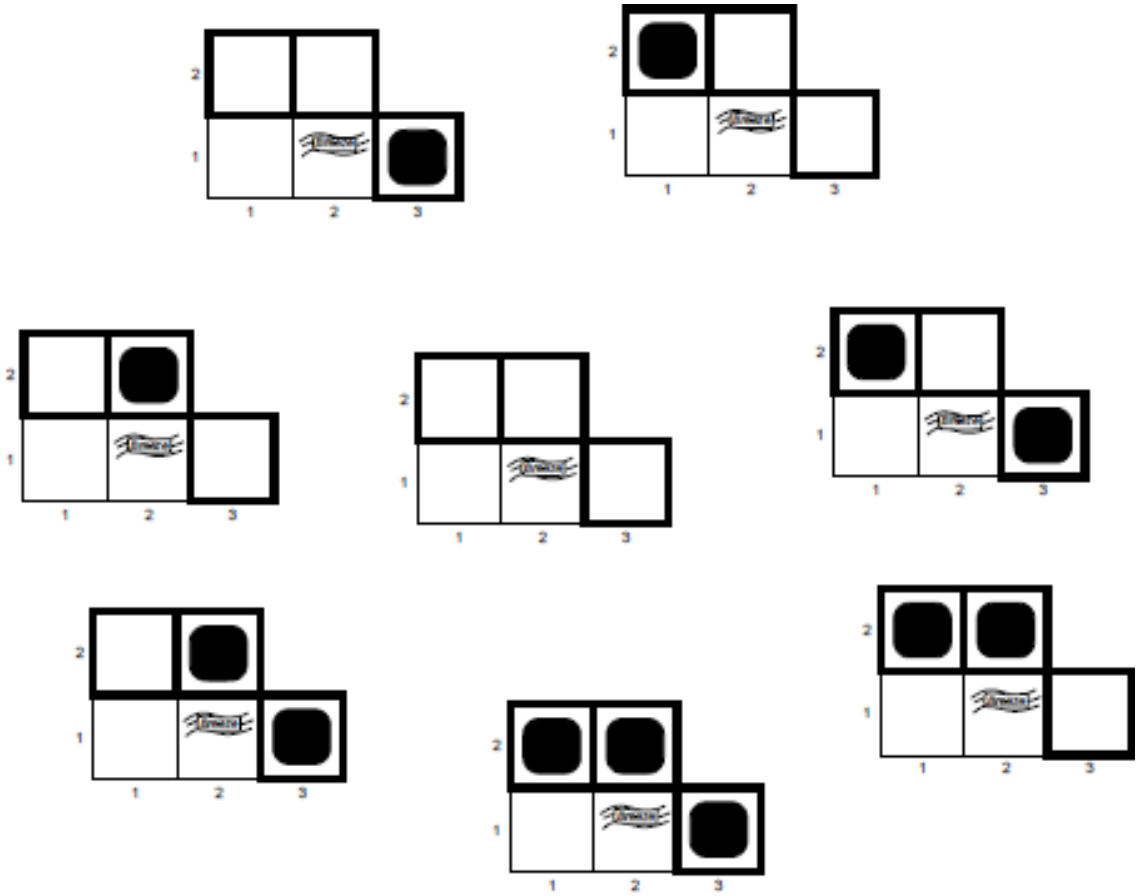
3 choices

8 possible models

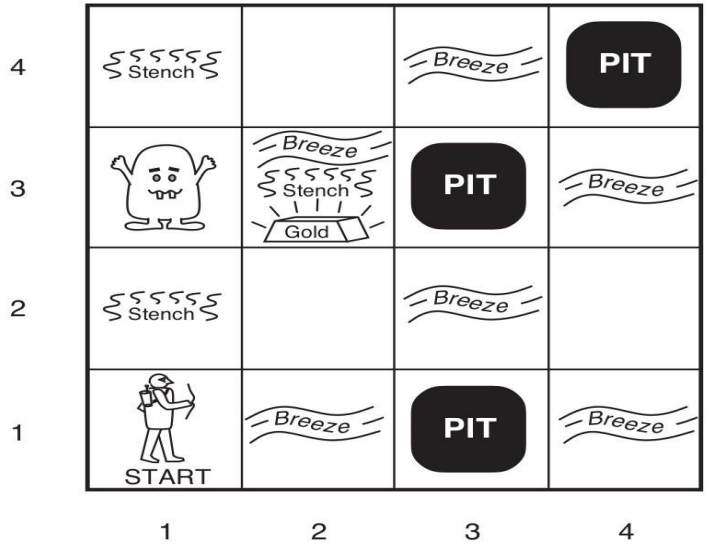


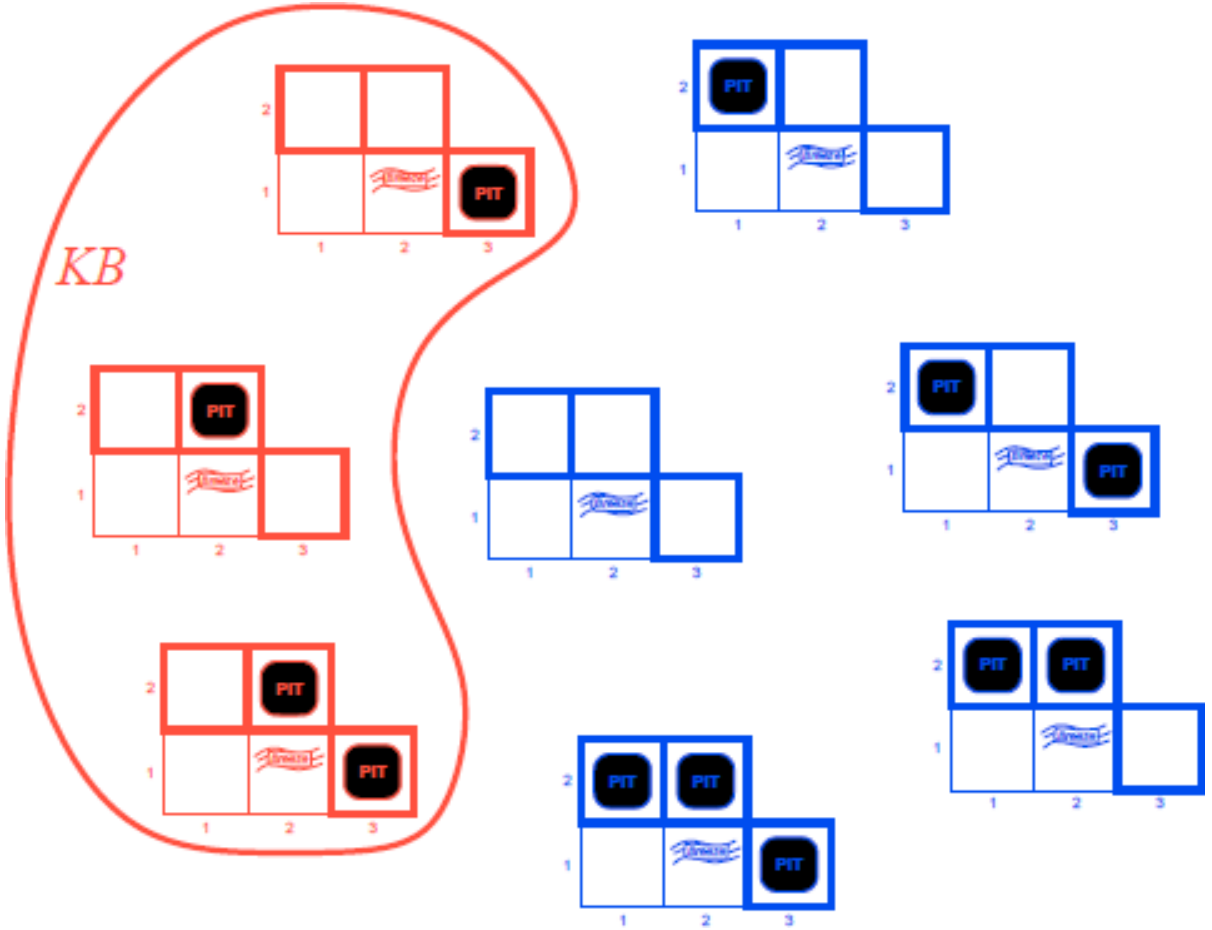
Entailment in Wumpus World: Wumpus Models





Wumpus Models: is [1,2] safe?





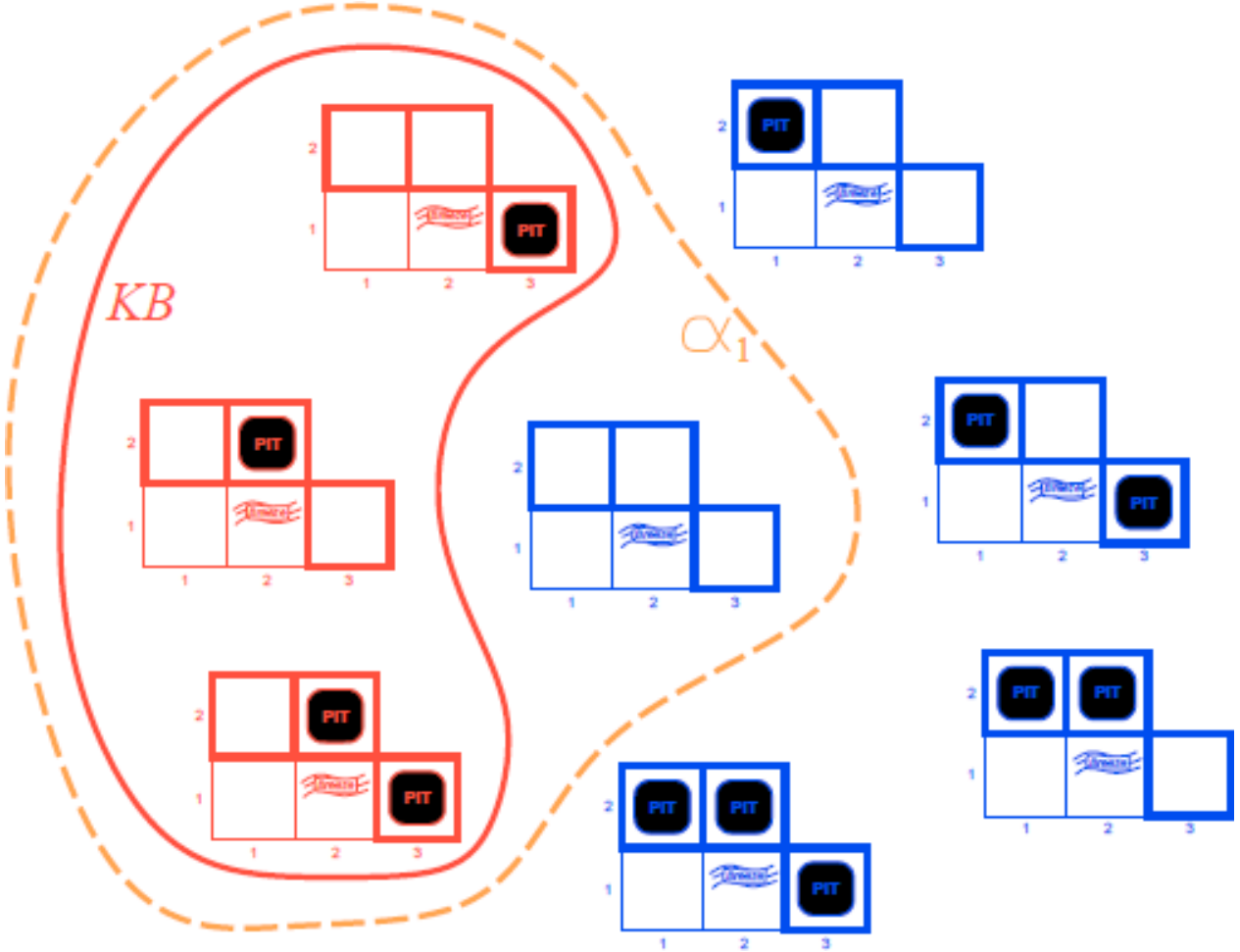
KB = wumpus-world rules + observations

Wumpus Models: is [1,2] safe?

KB = wumpus-world rules + observations

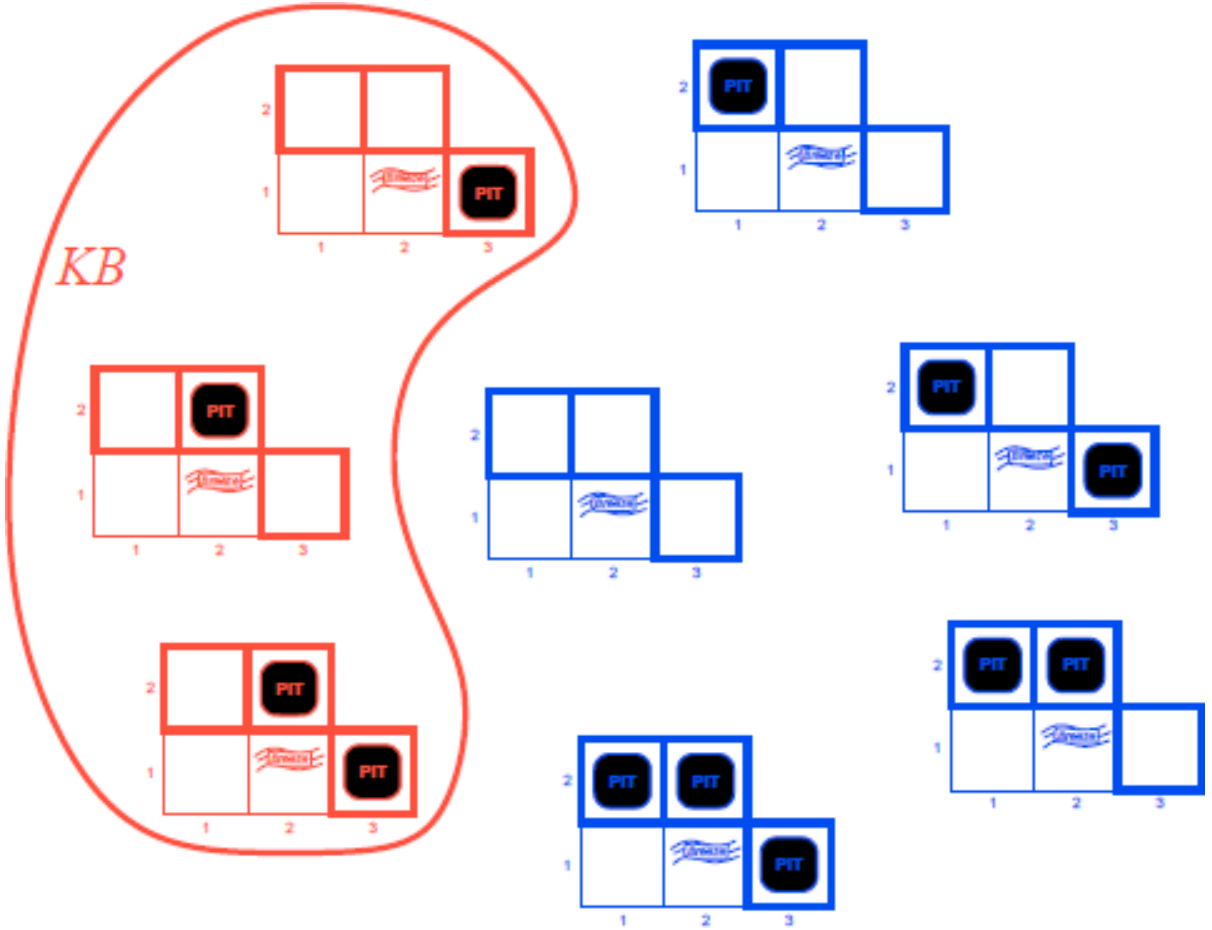
α_1 = "[1,2] is safe",

KB c α_1, proved by model checking

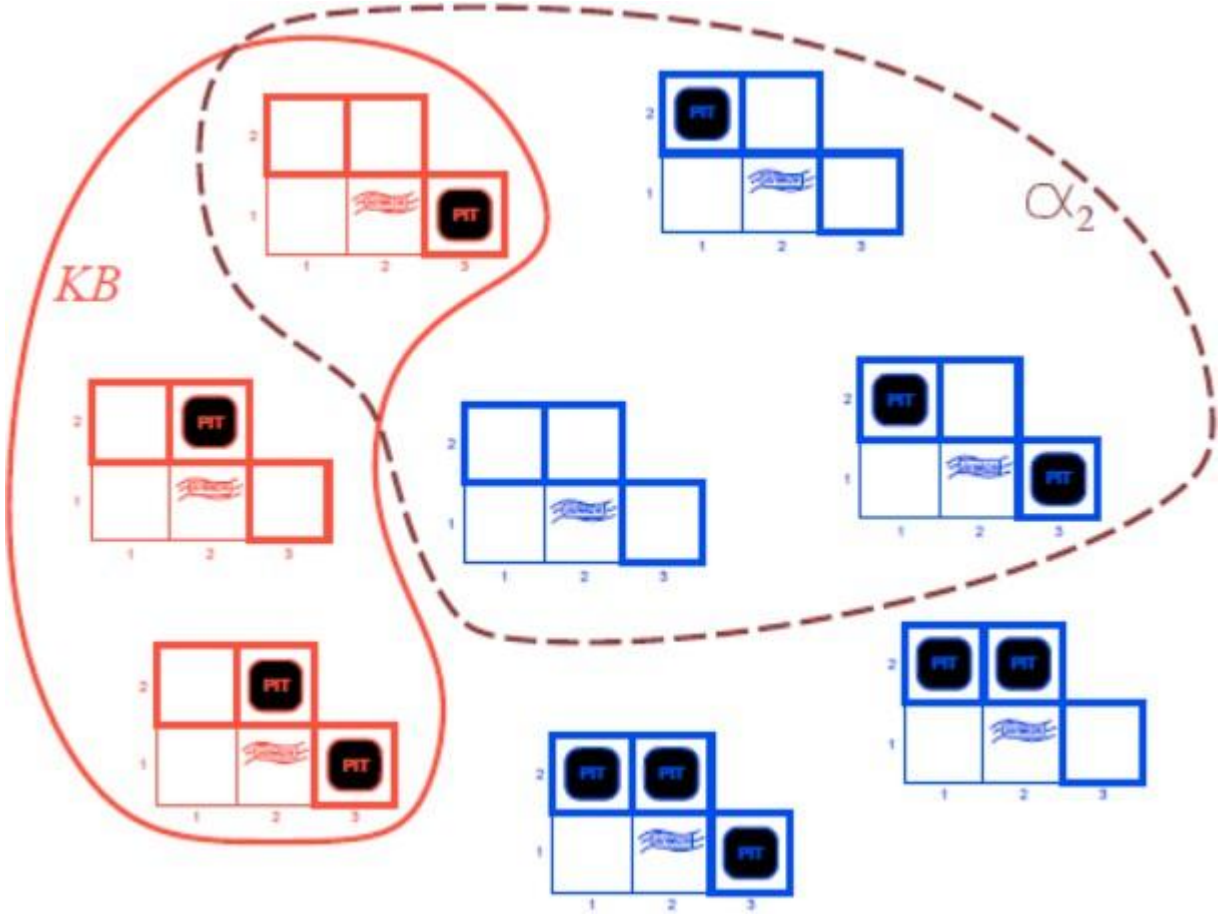


Wumpus Models: is [2,2] safe?

KB = wumpus-world rules + observations



Wumpus Models: is [2,2] safe?



KB = wumpus-world rules + observations

α_2 = "[2,2] is **NOT** safe",

KB $\not\models \alpha_2$, proved by model checking

- the agent cannot conclude that there is no pit in [2,2]. (Nor can it conclude that there is a pit in [2,2].